



Ammonium Alginate

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1 Nonproprietary Names

None adopted.

2 Synonyms

Alginic acid, ammonium salt; ammonium polymannuronate; E404; *Keltose*.

3 Chemical Name and CAS Registry Number

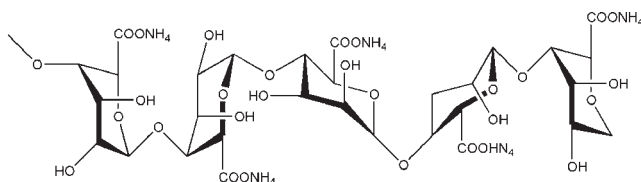
Ammonium alginate [9005-34-9]

4 Empirical Formula and Molecular Weight

(C₆H₁₁NO₆)_n 193.16 (calculated)
217 (actual, average)

Ammonium alginate is the ammonium salt of alginic acid.

5 Structural Formula



The number and sequence of the mannuronate and glucuronate residues shown above vary in the naturally occurring alginate. The associated water molecules are not shown.

6 Functional Category

Diluent; emulsifying agent; film-forming agent; humectant; stabilizing agent; thickening agent.

7 Applications in Pharmaceutical Formulation or Technology

Ammonium alginate is widely used in foods as a stabilizer, thickener and emulsifier. It is also used in pharmaceutical preparations as a color-diluent, emulsifier, film-former, and humectant.

8 Description

Ammonium alginate occurs as white to yellowish brown filamentous, grainy, granular, or powdered forms.

9 Pharmacopeial Specifications

See Section 18.

10 Typical Properties

Moisture content Not more than 15% at 105°C for 4 hours.

Solubility Dissolves slowly in water to form a viscous solution; insoluble in ethanol and in ether.

11 Stability and Storage Conditions

Ammonium alginate is a hygroscopic material, although it is stable if stored at low relative humidities and cool temperatures.

12 Incompatibilities

Incompatible with oxidizing agents and strong acids and alkalis.

13 Method of Manufacture

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14 Safety

Ammonium alginate is widely used in cosmetics and food products, and also in pharmaceutical formulations such as tablets. It is generally regarded as a nontoxic and nonirritant material, although excessive oral consumption may be harmful.

15 Handling Precautions

Observe normal precautions appropriate to the circumstances and quantity of the material handled. Eye protection, gloves, and a dust respirator are recommended.

16 Regulatory Status

GRAS listed. Accepted in Europe for use as a food additive. Included in the FDA Inactive Ingredients Database (oral, tablets).

17 Related Substances

Alginic acid; calcium alginate; potassium alginate; propylene glycol alginate; sodium alginate.

18 Comments

Alginates are commonly used in wound dressings.⁽¹⁾ Chitosan and alginates have been used together to produce sponges for use as wound dressings, or matrices for tissue engineering.⁽²⁾ Alginate microspheres have been produced by internal gelation using emulsification methods.⁽³⁾

Although not included in any pharmacopeias, a specification for ammonium alginate is contained in the Food Chemicals Codex (FCC), see Table I.

Table I: FCC specification for ammonium alginate.⁽⁴⁾

Test	FCC 6 ⁽⁴⁾
Identification	+
Arsenic	≤ 3 mg/kg
Ash	≤ 7.0% after drying
Lead	≤ 5 mg/kg
Loss on drying	≤ 15.0%
Assay	18.0–21.0% of CO ₂ , corresponding to 88.7–103.6% ammonium alginate

19 Specific References

- 1 Morgan D. Wounds—what should a dressing formulary include? *Hosp Pharm* 2002; 9(9): 261–266.
- 2 Lai HL *et al*. The preparation and characterization of drug-loaded alginate and chitosan sponges. *Int J Pharm* 2003; 251(1–2): 175–181.
- 3 Chan LW *et al*. Production of alginate microspheres by internal gelation using an emulsification method. *Int J Pharm* 2002; 242(1–2): 259–262.
- 4 *Food Chemicals Codex*, 6th edn. Bethesda, MD: United States Pharmacopeia, 2008; 44.

20 General References

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21 Authors

SA Shah, D Thassu.

22 Date of Revision

30 June 2008.



1 Nonproprietary Names

BP: Ammonium Chloride
PhEur: Ammonium Chloride
USP: Ammonium chloride

2 Synonyms

Ammonii chloridum; ammonium muriate; E510; sal ammoniac; salmiac.

3 Chemical Name and CAS Registry Number

Ammonium chloride [12125-02-9]

4 Empirical Formula and Molecular Weight

NH₄Cl 53.49

5 Structural Formula

See Section 4.

6 Functional Category

Acidifying agent; therapeutic agent.

7 Applications in Pharmaceutical Formulation or Technology

Ammonium chloride is used as an acidifying agent in oral formulations. It is also used as a food additive and antiseptic agent.⁽¹⁾

Ammonium chloride is used in the treatment of severe metabolic alkalosis to maintain the urine at an acid pH in the treatment of some urinary tract disorders or in forced acid diuresis.^(2–4) It is also used as an expectorant in cough medicines.⁽⁵⁾

8 Description

Ammonium chloride occurs as colorless, odorless crystals or crystal masses. It is a white, granular powder with a cooling, saline taste. It is hygroscopic and has a tendency to cake.

9 Pharmacopeial Specifications

See Table I.

Table I: Pharmacopeial specifications for ammonium chloride.

Test	PhEur 6.0	USP 32
Identification	+	+
Characters	+	—
Appearance of solution	+	—
Acidity or alkalinity	+	+
Loss on drying	≤ 1.0%	≤ 0.5%
Residue on ignition	—	≤ 0.1%
Thiocyanate	—	+
Bromides and iodides	+	—
Sulfates	≤ 150 ppm	—
Sulfated ash	≤ 0.1%	—
Calcium	≤ 200 ppm	—
Iron	≤ 20 ppm	—
Heavy metals	≤ 10 ppm	≤ 0.001%
Assay (dried basis)	99.0–100.5%	99.5–100.5%

10 Typical Properties

Acidity/alkalinity pH = 4.5–5.5 (5.5% w/w aqueous solutions at 25°C)

Density (bulk) 0.6–0.9 g/cm³

Hygroscopicity Hygroscopic with potential to cake.

Melting point Decomposes at 338°C; sublimes without melting.⁽⁶⁾

Solubility Soluble in water; hydrochloric acid and sodium chloride decrease its solubility in water. Also soluble in glycerin; sparingly soluble in methanol and ethanol. Almost insoluble in acetone, ether, and ethyl acetate.

Specific gravity 1.527 g/cm³

Vapor pressure 133.3 Pa (1 mmHg) at 160°C

11 Stability and Storage Conditions

Ammonium chloride is chemically stable. It decomposes completely at 338°C to form ammonia and hydrochloric acid. Store in airtight containers in a cool, dry place.

12 Incompatibilities

Ammonium chloride is incompatible with strong acids and strong bases. It reacts violently with ammonium nitrate and potassium chlorate, causing fire and explosion hazards. It also attacks copper and its compounds.

13 Method of Manufacture

Ammonium chloride is prepared commercially by reacting ammonia with hydrochloric acid.

14 Safety

Ammonium chloride is used in oral pharmaceutical formulations. The pure form of ammonium chloride is toxic by SC, IV, and IM routes, and moderately toxic by other routes. Potential symptoms of overexposure to fumes are irritation of eyes, skin, respiratory system: cough, dyspnea, and pulmonary sensitization.⁽⁷⁾ Ammonium salts are an irritant to the gastric mucosa and may induce nausea and vomiting.

LD₅₀ (mouse, IP): 1.44 g/kg⁽⁸⁾

LD₅₀ (mouse, oral): 1.3 g/kg

LD₅₀ (rat, IM): 0.03 g/kg⁽⁹⁾

LD₅₀ (rat, oral): 1.65 g/kg⁽¹⁰⁾

15 Handling Precautions

Observe normal precautions appropriate to the circumstances and quantity of the material handled.

All grades of ammonium chloride must be kept well away from nitrites and nitrates during transport and storage. They must be stored in a dry place, and effluent must not be discharged into the drains without prior treatment.

Ammonium chloride decomposes on heating, producing toxic and irritating fumes (nitrogen oxides, ammonia, and hydrogen chloride).

16 Regulatory Status

GRAS listed. Included in the FDA Inactive Ingredients Database (oral syrup, tablets). Accepted for use as a food additive in Europe. Included in medicines licensed in the UK (eye drops; oral syrup).